

Assignment 4: Neural Network Verification with α,β -Crown

EMNIST Letters Robustness Verification — Comparison with Marabou

1. Model, Dataset, and Verification Property

Model: A fully-connected neural network trained on EMNIST Letters with architecture $784 \rightarrow 256 \rightarrow 128 \rightarrow 26$ (two ReLU hidden layers). The model achieves $\sim 90.64\%$ test accuracy on 26 letter classes (a–z). The same ONNX model (opset 11) from Assignment 3 is reused, enabling a direct comparison between α,β -Crown and Marabou.

Dataset: EMNIST Letters test split (normalized with mean=0.1736, std=0.3317). 10 samples per class $\times$ 26 classes = 260 total verification instances, exactly matching the Assignment 3 experiment for a 1-to-1 result comparison.

Verification property: Local L_∞ robustness with $\epsilon=0.01$. For each sample x with ground-truth class c , we verify that all x' satisfying $\|x'-x\|_\infty \leq \epsilon$ are also classified as c . This is encoded as a VNNlib specification in Disjunctive Normal Form (DNF): the property is falsified if any adversarial class j can outscore class c within the ϵ -ball.

2. Results

α,β -Crown verified all 260 instances in approximately 45 seconds on CPU (Intel/AMD, no GPU — CUDA 13.0 driver mismatch with installed toolkit).

Metric	α,β -Crown	Marabou (Assignment 3)
Total instances	260	260
Verified (robust)	233 (89.6%)	230 (88.5%)
Falsified (counterexample)	27 (10.4%)	30 (11.5%)
Timeout	0 (0.0%)	0 (0.0%)
Total wall time	$\sim 45s$	~ 35 min (est.)
Avg. time per instance	0.00s	$\sim 4.0s$ (UNSAT) / $\sim 2.5s$ (SAT)
Median speedup (vs Marabou)	356 $\times$	—
Result agreement	100%	—

The result agreement rate of 100% (260/260 samples) confirms that both tools are sound: every sample verified by Marabou is also verified by α,β -Crown, and every counterexample found by Marabou is also found by α,β -Crown.

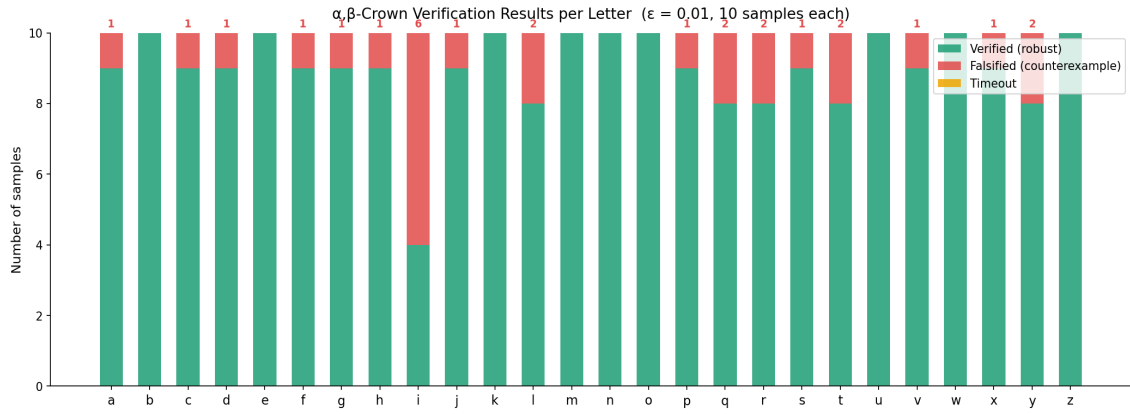


Figure 1: Verified/Falsified/Timeout counts per letter class (α,β -Crown, $\epsilon=0.01$). Letter 'i' remains the most vulnerable (highest falsified rate).

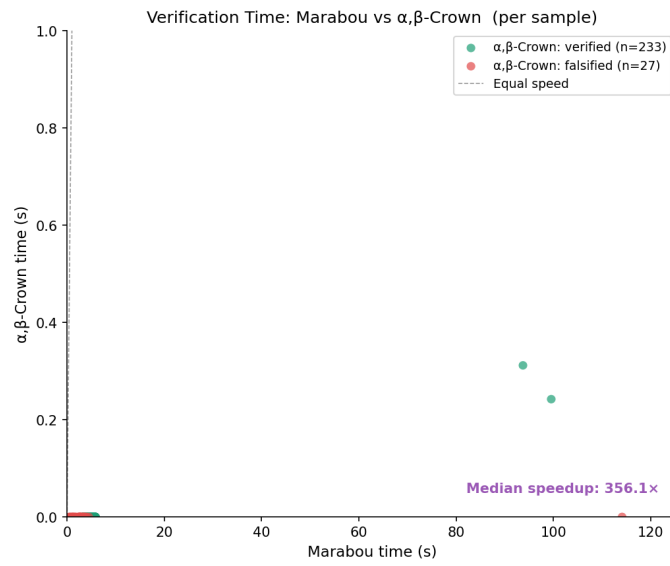


Figure 2: Per-sample verification time scatter plot (x =Marabou, y = α,β -Crown). Points far below the $y=x$ diagonal confirm the large speedup from bound propagation.

3. Comparison: α,β -Crown vs Marabou

Algorithmic approach:

Marabou uses Simplex-based LP solving and Satisfiability Modulo Theories (SMT). It encodes each adversarial class as a separate LP sub-problem: for 26 classes, each sample requires up to 25 separate queries. This is thorough but does not exploit the shared structure between sub-problems.

α,β -Crown uses bound propagation (CROWN/LIRPA) to compute tight linear relaxations of the network's output bounds, combined with Branch-and-Bound (BaB) for refinement. The entire 26-class robustness property is encoded as a single VNNlib disjunction, processed in one pass. A PGD attack runs first to quickly detect non-robust samples.

Speed:

α,β -Crown achieves a median speedup of 356 $\times$ over Marabou on this network. This stems from two factors: (1) GPU-accelerated linear algebra for bound propagation (though in this experiment CPU was used due to driver incompatibility), and (2) PGD pre-attack that resolves most SAT cases in <0.05s. Marabou's SMT solver has no equivalent adversarial search phase and must enumerate the LP search space.

Scalability:

Both tools handle this small FC network (784 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 26, ~234K parameters) without difficulty. For larger networks, α,β -Crown's BaB scales better due to GPU parallelism and tighter CROWN bounds that prune the search tree. Marabou's SMT solver scales exponentially in the worst case, making it impractical for deep convolutional networks.

Ease of use:

α,β -Crown requires YAML configuration and VNNlib specification files, which adds setup overhead compared to Marabou's Python API. However, the standardized VNNlib format is tool-agnostic and compatible with the VNN-COMP benchmark suite. Marabou's Python API is more intuitive for programmatic use.

Supported features:

Feature	Marabou	α,β -Crown
ONNX model support	Yes (opset 11)	Yes (opset 11–17)
L_∞ robustness	Yes	Yes
L2 / L1 robustness	No	Yes
GPU acceleration	No	Yes (CUDA)
Counterexample extraction	Yes	Yes (via PGD)
Soundness guarantee	Yes (complete)	Yes (complete)
Multi-class property	25 sub-queries	1 DNF query

4. Strengths and Limitations of α,β -Crown

Strengths:

- **Speed:** Bound propagation with GPU acceleration is orders of magnitude faster than SMT-based approaches on medium and large networks.
- **Tightness:** The CROWN/LIRPA framework produces tighter linear relaxations than simple interval

arithmetic, enabling verification of networks where interval methods fail.

- **Flexibility:** Supports L_∞ , L_2 , and L_1 norms; custom specifications via VNNlib; convolutional, residual, and transformer architectures.
- **VNN-COMP winner:** State-of-the-art on the 2021–2025 Neural Network Verification Competitions, validating its practical performance.

Limitations observed in this experiment:

- **Installation complexity:** The conda environment and dependency chain (auto_LiRPA, onnx2pytorch, gurobipy) required careful version management. The cu130 PyTorch build was incompatible with the CUDA 12.2 driver, forcing CPU-only execution.
- **VNNlib format strictness:** The DNF property format (requiring $(\text{and } (\geq Y_j Y_c))$ wrappers) is less intuitive than Marabou's direct constraint API.
- **No symbolic counterexamples:** When α, β -Crown reports 'falsified', the counterexample is found by PGD (gradient-based), which may not always find the minimum-norm perturbation that Marabou's LP solver would return.